Release Bundle

FirmwareSight 0.6.0 · release `release-e3b7d5c88b39095f23c8d057e8d257e09e40820598552fe4665e93e239d3bb23`

1. Release identity

Four different identities, named separately on purpose.

Release id	`release-e3b7d5c88b39095f23c8d057e8d257e09e40820598552fe4665e93e239d3bb23`
Project release version	`1.2.3` — resolved from `release.expected_version`
FirmwareSight app version	`0.6.0`
Snapshot analyzed	`snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6`
Analysis normalization rule	`p0-normalize-1`
Workspace HEAD observed by the Release Gate	`8bc3d25b99a5b739b42ad97d14041608212c91d1`
Workspace tag observed by the Release Gate	`v1.2.3`
Workspace dirty observed by the Release Gate	false

These Git facts describe the workspace the Release Gate ran in. They are not evidence about how the shipped bytes were compiled, and no claim of that kind is made here.

Release visual overview



Visual summary only. Detailed evidence follows. Browser Print / Save as PDF; native PDF export is unavailable.

2. Shipped artifacts

2 file(s) under `artifacts/`. The baseline build is never shipped: it exists only to explain the comparison in section 4.
The bytes this release consists of, in the order the release fingerprint lists them.

Bundle path	Kind	SHA-256	Bytes
`artifacts/firmware.elf`	elf	`4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374`	17464
`artifacts/firmware.map`	map	`a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6`	4664

3. Analysis summary

What was read off the build, per input file.

File	Format	Entry point	Build id
`firmware.elf`	Arm 32-bit little · `elf`	`0x8000049`	no .note.gnu.build-id section is present

File	Format	Entry point	Build id
`firmware.map`	Unknown unknown-bit unknown · `map`	a linker MAP records addresses, not an entry point	a linker MAP carries no build-id

Memory budgets

Absence is printed as absence. A budget that is not a number was not measured, and is never shown as zero.

Nonvolatile / load image	376 bytes (exact)
Runtime RAM	76 bytes (exact)
Accounting rule	`adr-0021-dual-budget`
Layout source	`map`
Weakest evidence basis	map-memory-configuration+elf-load
Admissible for a hard block	true
Device metadata held out of both budgets	4294 bytes

Coverage of this build

Rows carried in the analysis document	19 section(s), 52 symbol(s), 11 evidence record(s)
ELF support	`supported`
Sections	`available`
Symbols	`available`
Debug info	`available`
MAP companion	`provided`
Object attribution	`unavailable`
Git facts	`unknown`

4. Compare summary

Totals moved between the two builds. The full row-by-row comparison is in `diff.json`.

Base (old) snapshot	`snap-3f615b6247179d930f59b76dcdbaea1a5eca8c835ab6d410feb20a062f33f21f-p0-normalize-1-832690060a8d25f8acacd85a62ce76bc83dfb6768e435bcc04c51f9664ed363e`
Target (new) snapshot	`snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6`
Nonvolatile / load image	256 bytes → 376 bytes · delta +120 (exact)
Runtime RAM	8 bytes → 76 bytes · delta +68 (exact)
Comparability of the pair	exact
Rows that differ	sections: 1 added, 1 removed, 16 changed, 0 unpaired · symbols: 38 added, 33 removed, 5 changed, 67 unpaired
Identical, so left out of the tables	2 section(s), 9 symbol(s)

Top growth, as Core ranked it

Top section growth

Delta	Row	Bytes	Change
+221	`.debug_loclists`	414	Changed
+112	`.debug_info`	910	Changed
+107	`.debug_line`	421	Changed
+88	`.text`	212	Changed
+80	`.symtab`	848	Changed

Top symbol growth

Delta	Row	Bytes	Change
+42	`mqtt_task`	46	Changed
+16	`payload_extra`	20	Changed

The complete comparison ships as `diff.json` under schema `urn:firmwaresight:schema:diff:1` version 1.

5. Release Gate

✓ PASS — the disposition that made this bundle possible. A bundle is written only for a **PASS** after accepted reviews; a **REVIEW** or a **BLOCK** produces no release, and an evidence gap never becomes a review.

Run id	`gate-e8c307e670d73a059eeb62145a3cfd5024ca7c67dc2fc2c2689ecc0bb92f88c1`
Snapshot	`snap-4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374-p0-normalize-1-a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6`
Baseline	`snap-3f615b6247179d930f59b76dcdbaea1a5eca8c835ab6d410feb20a062f33f21f-p0-normalize-1-832690060a8d25f8acacd85a62ce76bc83dfb6768e435bcc04c51f9664ed363e`
Policy digest	`f65f9c921158ee615c5efcccdacd30106b4897dab1257a83b70311ae66609bab`
Project config schema	`1`
Severity before reviews	✓ PASS
Severity after accepted reviews	✓ PASS

Findings by state

Core's counts for this run, carried rather than recounted.

State	Count
✕ BLOCK	0
▲ REVIEW	0
? UNKNOWN	0
✓ PASS	6
– N/A	4

All findings

In the order a release owner reads them: what blocks, what needs a decision, what is unknown, what passed. Every finding of the run is listed, not a truncated top list.

Rule	State	Effective severity	Finding	Evidence	Next step
`release.version_matches_policy`	✓ PASS	same as state	Workspace tag `v1.2.3` matches the version pattern and its version `1.2.3` is the declared release version.	`policy:version.source` · `policy:version.pattern` · `git:exact-tag=v1.2.3` · `policy:release.expected_version`	none recorded
`artifacts.required`	✓ PASS	same as state	Every required artifact kind is present in the snapshot (elf, map).	`policy:artifacts.required=elf,map` · `artifact:elf:4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374` · `artifact:map:a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6`	none recorded
`artifacts.hashes`	✓ PASS	same as state	All 2 snapshot artifact(s) carry a SHA-256 digest.	`artifact:elf:4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374` · `artifact:map:a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6` · `policy:artifacts.hashes`	none recorded

Rule	State	Effective severity	Finding	Evidence	Next step
`memory.flash_budget`	✓ PASS	same as state	Nonvolatile footprint 376 B is within the 512 B budget (136 B headroom).	`policy:memory.flash_budget=512` · `evidence:ev-memory-nonvolatile_image_footprint_bytes`	none recorded
`memory.ram_budget`	✓ PASS	same as state	Runtime RAM footprint 76 B is within the 256 B budget (180 B headroom).	`policy:memory.ram_budget=256` · `evidence:ev-memory-runtime_ram_footprint_bytes`	none recorded
`release.notes`	✓ PASS	same as state	Release Notes `RELEASE_NOTES.md` is present.	`policy:release.require_release_notes` · `file:RELEASE_NOTES.md` · `file:RELEASE_NOTES.md#sha256=1169e4af5e3daec77ab9b6967e24436c704286e2638468f9efc2ea8ca83a8414`	none recorded
`git.clean`	– N/A	✓ PASS	`release.require_clean_git` is false, so a clean workspace is not required.	`policy:release.require_clean_git`	none recorded
`release.commit_matches_expected`	– N/A	✓ PASS	No `release.expected_commit` is declared for this release.	`policy:release.expected_commit`	none recorded

Rule	State	Effective severity	Finding	Evidence	Next step
`diff.growth`	– N/A	✓ PASS	No `[diff]` growth threshold is configured.	`policy:diff.flash_growth_review_bytes` · `policy:diff.ram_growth_review_bytes`	none recorded
`evidence.unknown_review`	– N/A	✓ PASS	`gate.unknown_evidence_review_count` is not configured.	`policy:gate.unknown_evidence_review_count`	none recorded

6. Accepted reviews

No review was accepted for this run, so the disposition in section 5 is the one Core computed with nothing set aside. An empty list is a fact, not a missing export.

7. Release notes

Included	yes, as `release-notes.md`
SHA-256 of the shipped copy	`1169e4af5e3daec77ab9b6967e24436c704286e2638468f9efc2ea8ca83a8414`
Bytes	51

The notes ship as their own file and are not reproduced inside this report. Markdown a reader wrote is never parsed, rewritten or embedded as raw HTML by the bundle, which is what keeps this document free of markup a note could carry.

8. Bundle integrity

Two layers of SHA-256 lists cover this bundle. `SHA256SUMS` hashes every payload file except itself and the manifest. `release-manifest.json` hashes every file except itself, that list included.
Every file a digest can be named for here. This report is payload: it is written before both index files, so it cannot print their digests, or its own.

Bundle path	SHA-256	Bytes
`accepted-reviews.json`	`155d39d698a4ef3239cdbc29c93ccef651a1994f44f3ac78e01b3ed0295f4109`	170
`analysis.json`	`ba20d979db4c92fd98780f8afc5186e0e51d7a8dfef8f40eb8ad6c3786bf199a`	26963

Bundle path	SHA-256	Bytes
artifacts/firmware.elf	4b4087e1407ceddcb1e1cfcd978eedbc8906cf88083ba3b4672c7c225ddd8374	17464
artifacts/firmware.map	a38575cd51ec2730442a0f3b8506c99fc49eec68d4b28ecdc9dda9434709ace6	4664
diff.json	1930cdb6a098c67456297ec321910315d21aeb3ce79ef7264fcc4037e148242a	64355
gate-results.json	329a8ee37ab3eb16843803fd024553df9c82e4a7f2c501647ac7e0122a198e55	4501
release-notes.md	1169e4af5e3daec77ab9b6967e24436c704286e2638468f9efc2ea8ca83a8414	51
release-report.html	indexed by release-manifest.json, which is written after this report	
SHA256SUMS	indexed by release-manifest.json, which is written after this report	
release-manifest.json	indexed by release-manifest.json, which is written after this report	

The self-reference rule

SHA256SUMS covers	every bundle file except itself and the manifest
SHA256SUMS excludes	SHA256SUMS, release-manifest.json
release-manifest.json covers	every bundle file except itself, SHA256SUMS included
release-manifest.json excludes	release-manifest.json

A file cannot contain its own cryptographic hash without a fixed-point scheme, so no self-digest is faked: self_digest_written is false. no file carries its own digest; SHA256SUMS hashes the payload, release-manifest.json hashes the payload plus SHA256SUMS

Verifying this bundle without FirmwareSight

Every file here is a plain file. Any SHA-256 tool and any text editor is enough. The check below proves that the bytes match the lists — an integrity and consistency statement about the bundle as written. It says nothing about who wrote it, and nothing in this bundle is a signature.

1. List the directory. It must contain exactly the files release-manifest.json indexes, plus that manifest itself, and no temporary or hidden file.
2. Recompute the SHA-256 of every file named in SHA256SUMS and compare it with the digest on that line. On a POSIX host one command does the whole list: sha256sum -c SHA256SUMS.

3. On Windows, `Get-FileHash -Algorithm SHA256 <path>` prints the same digest in upper case; lower-case it before comparing, because both lists are written in lowercase hex.
4. Compare every line of `SHA256SUMS` with the entry of the same path in the `files[]` array of `release-manifest.json`. Digests and sizes must agree, and the manifest must additionally carry an entry for `SHA256SUMS` itself.
5. Confirm that neither list contains a blank digest, an all-zero digest, or the digest of the file it sits in. There should be none of those, and looking for them is what makes the rule in the table above checkable rather than a claim.
6. Open `analysis.json`, `gate-results.json` and `accepted-reviews.json` as plain JSON. Each names the schema it is an instance of, and the manifest's `extensions` block states which schema majors this bundle was written against.

9. Known capability limits of this release

What the inputs of this build did not allow FirmwareSight to establish. A limit is stated as a limit instead of being left as an empty cell.
Capability reporting, taken from the analysis document in this bundle.

Capability	Reported	What that means for this release
Object attribution	`unavailable`	Totals are reported per section and per symbol only. No per-object-file or per-module total is claimed here, and none should be inferred from the section rows.
Git facts	`unknown`	This capability is reported at less than full coverage by the analysis document in this bundle.

Object and module attribution

the persisted symbol and section records carry no object-file or module attribution, so no object or module delta is claimable for either build

Comparison warnings

- `SYMBOL-AMBIGUOUS` — 67 symbol row(s) repeat a name, kind and binding combination on one or both sides and were left unpaired; a row position is not an identity

Deterministic export: no timestamp, no scripts, no network requests. Written by FirmwareSight 0.6.0 over the documents in this bundle.